SWAPNALI JADHAV

AI/ML Engineering Student
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EDUCATION

Smt. Indira Gandhi College of Engineering, Mumbai University 2023 – 2027
B.E., Artificial Intelligence & Machine Learning (AIML) | CGPA: 8.5 / 10 (Third Year – Overall)

SKILLS

- **Languages:** Python, Java
- **Tools & Frameworks:** OpenCV, MediaPipe, TensorFlow
- **Concepts:** Data Structures & Algorithms, Machine Learning, LLMs, DBMS, OOP

PROJECTS

AEGIS – Explainable Spacecraft-Health Monitoring & Decision Support | *Python, Machine Learning, Anomaly Detection, Simulation* | *Winning Project, Hackathon 2026*

- Built a spacecraft-health monitoring prototype that turns telemetry (battery voltage, temperature, wheel speed, power) into explainable fault alerts and safety-checked response recommendations through a detect → explain → recommend → safety-check workflow.
- Added telemetry-quality checks (missing, stale, noisy data) so sensor/data issues are not mistaken for hardware faults; combined statistical and ML detection over multi-channel patterns; ranked likely root causes with matched and contradicting evidence.
- Separated detection from action approval using a rule-based guard over a predefined action library; built a simulated spacecraft with fault injection. In a saved simulation benchmark, detected 29 of 40 injected fault episodes (72.5%), including 5 of 5 reaction-wheel friction episodes.

Air-Hand Gesture Recognition System | *Python, OpenCV, MediaPipe*

- Built a contactless interaction system using sensors, cameras, and MediaPipe-based tracking to translate in-air hand movements into device actions, enabling hygienic, hands-free control for smart devices, VR, and in-car/gaming systems; presented at a national-level paper conference.
- Delivered four touch-free modules: Virtual Painting, Virtual Mouse, Virtual Volume Controller, Virtual Keyboard.

Plant Whisper – AI-Powered Plant Monitoring | *Python, ML, IoT Sensors* | *(IRJET Published Paper)*

- Conducted interdisciplinary research studying how plants perceive and respond to their environment via biochemical and electrical signals.
- Integrated environmental sensors, real-time data acquisition, and ML to monitor plant responses to light, temperature, soil moisture, sound, and human interaction, building predictive models for plant health with applications in precision farming and smart gardening.

Kisan GPT – Multilingual AI Agricultural Assistant | *Python, Hugging Face Transformers, LangChain, Streamlit*

- Built an AI conversational assistant delivering crop advisory, fertilizer/pest guidance, real-time weather updates, and government scheme information to farmers in Hindi, Marathi, and English via text or voice.
- Integrated translation and speech APIs to enable seamless multilingual, multimodal interaction.

AI Legal Document Checker | *Python, LLMs (GPT/BERT), NER, RAG*

- Built an NLP-based system that reviews legal documents for grammatical errors, ambiguous/missing clauses, and compliance issues against standard templates.
- Combined LLMs, Named Entity Recognition, text classification, and RAG to flag risks and generate correction suggestions, reducing manual review effort.

WORK EXPERIENCE

Intern – Large Language Models (LLMs), College Research Internship May 2025 – July 2025

- Set up and deployed a Ray-based LLM benchmarking environment using Locust.
- Experimented with LLaMA-2 integration and fine-tuned model inference for chat completion endpoints.
- Worked with llm-on-ray configurations and optimized compute nodes for distributed model serving.

Outreach Intern, Scroll Connect February 2025 – April 2025

- Created Gen-Z focused tech + culture content, increasing engagement by 35%.
- Researched AI trends and simplified complex concepts for digital storytelling.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Co-Lead**, SparksCode (College Technical Club) – coordinate technical activities and drive member engagement.
- **AWS Campus Ambassador** (Current) – represent AWS on campus, driving student engagement with cloud technologies.
- **Vice President**, Research & Development Cell, AIML Department – led departmental R&D initiatives, including AI in Agriculture & Healthcare projects, and coordinated technical activities.
- Active college volunteer and Event Head for multiple technical and cultural fests – managing planning, coordination, and execution.
- Anchored and hosted college events and technical symposiums, strengthening public speaking and stage presence.

ACHIEVEMENTS

- **Winner, Hackathon 2026** – built AEGIS, an explainable spacecraft-health monitoring prototype that detects faults, ranks likely causes, and safety-checks recommended responses.
- Shortlisted for Smart India Hackathon 2024 (Smart Education theme).
- Built a Smart Waste Segregation System using real-time ML object detection.